

Lab 7: Instrumental variable analysis

Causal Machine Learning

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The dataset for this practical is `igf.dta`. These data are from a study of 139 premenopausal women for whom data are available on total area of dense tissue in the breast, serum IGFBP-3 levels and also on a putative functional gene for this protein, IGFBP3g which has genotypes AA, AC and CC. The motivation for these data is the study of the relation between growth hormones and physiological features of the breast although there is concern that any association may be confounded by several other breast cancer risk factors, such as reproductive behaviour.

1. Examine the association of lucent area (or better its log-transformation) with genotype.
2. Examine the histogram of serum IGFBP-3, the exposure of interest. Again consider log-transforming this variable to facilitate the examination of its association with other variables. Summarise its distribution by genotype. Could the genotype act as an instrument for serum IGFBP-3? Check whichever conditions for this are testable from the data (if any).
3. Draw a DAG depicting what you think are the plausible causal association between the outcome `log(lucent)`, the exposure of interest, serum IGFBP-3 `log(igfbp3)`, and the potential confounders `age`, age at first birth (`afbc`), breast feeding duration (`brfcat`), number of children (`parcat`) and ever having taken oral contraceptives (`evpil`). Add the instrument and potential unobserved confounders. What assumptions are required for the genotype to be an instrumental variable for the effect of `ln(IGFBP-3)` on `ln(lucent area)`?
4. Use an instrumental variable analysis to estimate the effect of `ln(IGFBP-3)` on `ln(lucent area)`, using IGFBP3g as an instrumental variable. Calculate an accompanying standard error based on the influence function.
5. A possible issue with the IV estimates above is that the exclusion restriction assumption may not be appropriate. Indeed parity and age at first birth for example could be on the causal pathway from genotype to outcome (as the gene could influence fertility). We could therefore consider blocking this pathway by controlling for age at first birth. Using debiased machine learning to estimate the resulting effect of `ln(IGFBP-3)` on `ln(lucent area)`, along with a 95% confidence interval. Given the small size of the dataset, we will restrict to ‘simple’ data-adaptive methods and no cross-fitting.
6. If time allows, then compare your results with those obtained using assumption-lean linear regression for the association between the outcome association between the outcome `log(lucent)`, the exposure of interest, serum IGFBP-3 `log(igfbp3)`, and the potential confounders `age`, age at first birth (`afbc`), breast feeding duration (`brfcat`), number of children (`parcat`) and ever having taken oral contraceptives (`evpil`).

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